THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 2: **Electricity: Circuits and their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which choice correctly lists materials in order: conductor, insulator, conductor? 1

(a) rubber, iron, plastic **(b)** plastic, copper, wood

(c) iron nail, rubber, aluminium foil **(d)** wood, glass, copper

2. Which effect of electric current is utilised in a fuse? 1

(a) heating effect **(b)** light effect **(c)** sound effect **(d)** chemical effect

3. In an electric kettle, the heating coil is immersed in water yet is insulated by: 1

(a) air

(b) plastic only

(c) nothing

(d) a sealed metal sheath with insulating powder

4. Why is a thick wire used for heavy appliances like geysers? 1

(a) it is cheaper

(b) it does not conduct

(c) it can carry more current without overheating

(d) it is lighter

5. Which of the following is a reason for coiling the heating wire in an electric heater? 1

(a) coiled wire is an insulator **(b)** to cool the wire

(c) a long wire fits in a small space **(d)** to stop current

6. Which of the following may reduce the electricity bill? 1

(a) switching to LED bulbs **(b)** using old heaters

(c) leaving lights on **(d)** using many bulbs

7. Why are electrical sockets kept away from water in kitchens and bathrooms? 1

(a) water makes wires thick **(b)** water is an insulator

(c) water makes sockets cold **(d)** water conducts and can give a shock

8. Which of the following might cause overloading? 1

(a) connecting many appliances to one socket **(b)** switching off

(c) using a fuse **(d)** using one bulb

9. **Assertion (A):** In a series connection of bulbs, if one bulb is removed the other bulbs go off. 1

Reason (R): In a series connection there is only one path for the current.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** An MCB switches off the circuit when the current becomes too large. 1
Reason (R): An MCB can be switched on again and reused, unlike a fuse wire.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a bulb in a circuit glow immediately when the switch is on? 2

12. How are cells connected to increase the voltage? Show with an example. 2

13. Why is it important that wires of the right thickness are used in the house wiring? 2

14. Explain why metals are used for making heating plates in toasters. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how a short circuit can start a fire. 3

16. Why is distilled water not used for the tester but tap water works? 3

17. Describe why wires used for transmission are made of aluminium and not copper in many places. 3

OR

Describe a method to identify the conductors among liquids like lemon juice, sugar solution and salt solution. 3

18. Explain why a current of 4 A makes a wire hotter than 1 A in the same wire. 3

OR

Why must we not use a bulb of higher wattage than the holder's rating? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** For Diwali, Ravi buys a string of 20 small bulbs joined one after another. When one bulb fuses, the entire string goes off. The shopkeeper says that in house wiring the bulbs and fans are connected in parallel. 5
(i) What type of connection is used in Ravi's string?
(ii) Why do all the bulbs go off when one fuses?
(iii) What would happen to the other bulbs if they were in parallel and one fused?
(iv) Give two reasons why appliances in a house are connected in parallel.

20. Discuss the working of a simple tester that can be made with a cell, a bulb and wires, and its use in finding conductors, including limitations. 5

OR

Explain how a fuse is connected in a circuit and why. 5

END OF QUESTION PAPER — Total: 40 Marks